

SENG1050

Lab 9

In this lab, you will gain experience in good web design and HCI practices.

[Task 1](#) - [Task 2*](#) - [Task 3](#) - [Task 4](#)

Assessment 3 Milestone 2 (Important!)

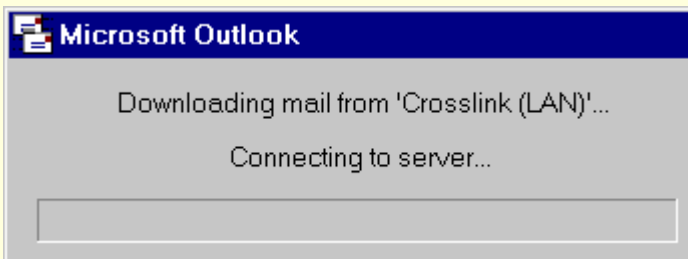
Approaching this week, you should have completed milestone 1 of Assessment 3 (Web-Based Assignment 2). Time yourselves and try to complete milestone 2 of Assessment 3 by Week 11. Please refer to the Assessment 3 documentation and follow the steps listed carefully.

Upon completion of milestone 2, approach your lab instructor to claim another 1% of your Assessment 3 score. Once you have been marked off for the milestone, continue the tasks below.

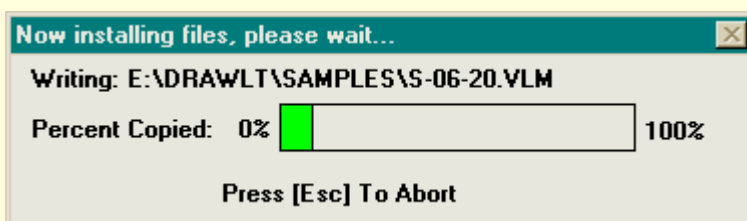
Task 1 - User Interface (Optional)

Discuss how you can improve the following problems based on the information in the [Interface Hall of Shame](#).

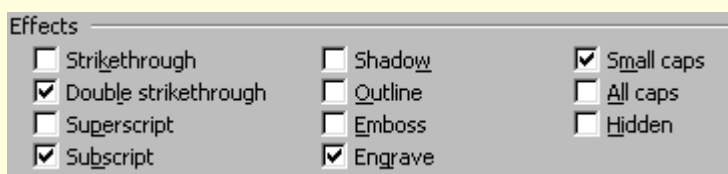
1. Is there progress?



or



2. I want them all!



3. Fun with scrolling!

Human Resource Administration

Name: SMITH, JOHN Q

Status: Payroll: G

Category: C

Sequence #: 5

Awards

Mgr ID	Date	Amount

or

Customer Information

First Name: Last Name: Middle:

Address1: Address2:

City: State: Zip:

Account Number: Mother's Maiden Name:

4. More tabs please!

Customize

User interface | Program execution | Spelling

Tagging | File compare | Fonts | Sessions

Customize | Editing | Windowing | Files | Blocks

Projects | Weblair | VCS | Search

Find | Replace | File find | File replace | Hilite | Options

Type: ☐ Literal

Options: ☐ Backward search

or

Customize

Category:

- Blocks
- Customize
- Editing
- File Compare
- Files
- Fonts
- Program Execution
- Projects
- Search
- Sessions
- Spelling
- Tagging
- User Interface
- VCS
- Weblair
- Windowing

Page Break string: |12

Word delimits: .()".#\$@!%&^*()[]?/|.: |255|9

Max undo count: 100

Options

- ☒ Truncate spaces
- ☐ Control-Z at EOF
- ☒ CR/LF at EOF
- ☒ Edit binary files in hex
- ☐ Restrict cursor
- ☒ Lock cursor on v-scroll

Tab expand

- ☐ Tabs
- ☒ Spaces

Default mode

- ☒ Insert
- ☐ Overwrite

Paragraphs separated by

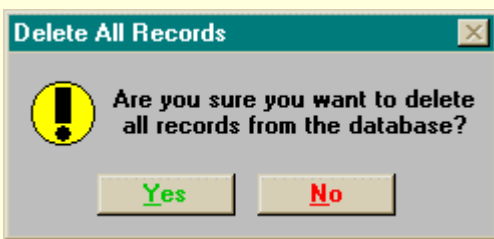
- ☐ Blank lines only
- ☒ Indent levels

OK Cancel Help

5. Helpful tips!



6. Green for the go!



Task 2 - Media Queries

Media queries can be used to ensure that CSS rules only apply to certain clients (e.g., desktop, mobile, etc.). We will look at some simple media queries using the provided *form-validation.html* from Canvas:

1. Create a *style.css* file and link it to the HTML document using the *link* tag.
2. In the stylesheet, add the following code:

```
@media (max-width: 600px) {body {color: yellow;}}
@media (min-width: 601px) {body {color: green;}}
```
3. Adjust the width of the browser and watch how the text colour changes depending on which media query evaluates to true.
4. Write a media query to make the text size larger on mobile devices (*Hint*: Use `@media (hover: none)` to target mobile devices.) Through the Inspector, check whether your query works by using the 'Toggle device emulation' button (second button on top left) to switch to mobile phone emulation.
5. Write a media query to change the background colour when the viewport width is less than the viewport height (i.e., portrait view). *Hint*: Use `min-aspect-ratio` or `max-aspect-ratio` in your query.

Task 3 - Stress Test (Optional)

An important part of HCI testing is to have other people look at your design, as what seems obvious to you may not be so obvious to others.

1. Take a look at the [navigation stress test](#) for websites. Look at the Hockey example at the bottom of the page and note the parts of the site that could be improved based on the results of the stress test.
2. Now try the stress test on a [different website](#). Go through the stress test questions and discuss the results. What questions are hard to answer? How could this page be improved?

Task 4 - Lighthouse Analysis (Optional)

Chromium-based browsers (e.g., Microsoft Edge and Google Chrome) have an additional tool in their Inspector called 'Lighthouse', which can check many of the common speed, search, and usability issues with websites.

1. Open a Chromium-based browser with extensions disabled (so they don't impact the results of the test). You can do this by opening the browser with the flag `--disable-extensions`. For example, Microsoft Edge can be launched using `msedge.exe --disable-extensions`.
 2. Open the HTML document in Task 2 in this new browser window. Open the Inspector and select the 'Lighthouse' tab.
 3. Run a scan in both Desktop and Mobile modes over all categories (except Progressive Web App). Take note of the results.
 4. Make adjustments to the page and try to increase the score in each category to 100% as a practice. Note that for our course, you may ignore the HTTPS and CSP issues under the 'Best practices' category.
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